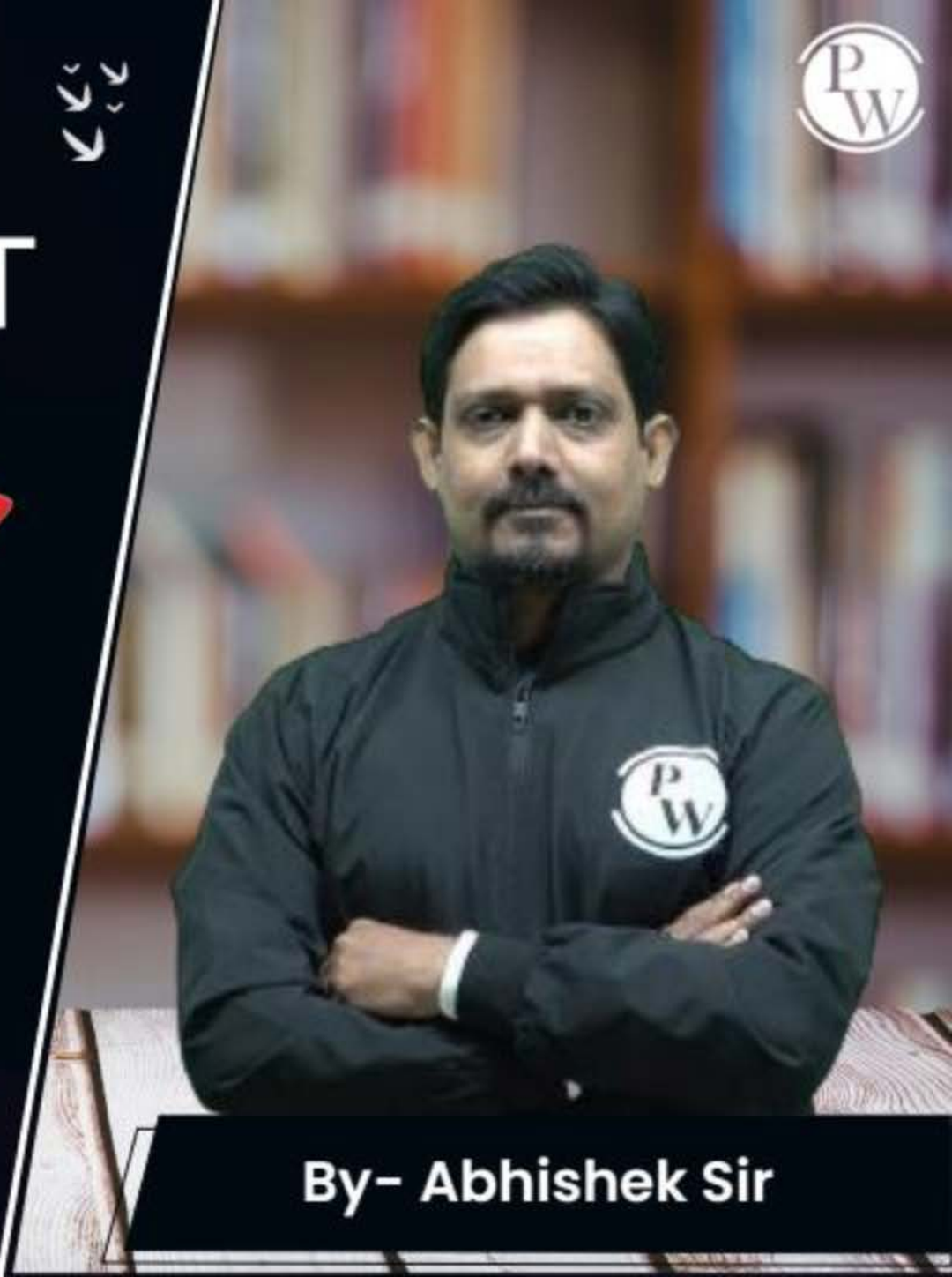
Computer Science & IT



C programming

Control Flow Statement

Lecture No. 03



By- Abhishek Sir

Recap of Previous Lecture



- Topic — Switch statement
- Topic — for loop
- Topic
- Topic
- Topic

Slide

Topics to be Covered



- Topic
- Topic
- Topic
- Topic
- Topic

Mathematical aspect

Nested for loop

Slide



For loop

- Decrement loop
- Other than ++ (i+2) formula
- Multiplication formula
- Nested for loop
- While loop
  - Flow chart
  - For loop equivalent to while loop



For loop: Syntax

Different range of numbers

```
for(i =1;i<=10;i++)
    printf("Name");
```

Slide



For loop: Syntax

Different range of numbers

```
for(i =11; i<=21; i++)
     printf("Name");
```



For loop: Syntax

Different range of numbers

```
for(i =-10; i<=21; i++)
    printf("Name");
```

LB < uB

uB - LB + 1



For loop: Syntax

Different range of numbers

```
for(i =-10; i<=21; i++)
    printf("Name");
```

21 − (−10) + 1 = 32

Slide



For loop: Syntax

Different range of numbers

LB :- -10 20

```
for(i =-10; i<21; i++)

    printf("Name");
```

20 - (-10) + 1 = 31

i++ → UB - LB + 1

Slide



For loop: Syntax

Different range of numbers

```
for(i =100; i>=1; i--)
    printf("Name");
```

Decremental for loop

UB-LB+1

UB = 100

LB = 1

UB-LB+1 = 100-1+1 = 100

Slide



For loop: Syntax

Different range of numbers

```
for(i =100; i>=1; i--)
    printf("Name");
```

Decremental for loop

UB-LB+1

UB = 100

LB = 1

UB-LB+1 = 100-1+1 = 100

```
i--;
--i;
```
} Same if Single use



For loop: Syntax

All Three Parts are mandatory

```
int i=1;
for( ; i<=10; i++)
    printf("Name");
```

None of the part is mandatory

10 time 10

if Not initialize then garbage value

Slide



For loop: Syntax

All Three Parts are mandatory

```
int i;
for( i=1; ; i++)
    printf("Name");
```

if condition is Not present

then it always evaluates to true

Infinite loop



For loop: Syntax

All Three Parts are mandatory

```
int i;
for( i=1;i<=10 ;)
     printf("Name");
```

i++ X

No

1<=10 true

1<=10 true

Name

Infinite loop

Slide



For loop: Syntax

All Three Parts are mandatory

```
int i;
for( ; ;)
    printf("Name");
```

if No condition is present

its always true

Infinite loop - : Name

Slide



For loop: Syntax

All Three Parts are mandatory

```
int i;
for( i=1;i<=10 ;i++);
     printf("Name");
```



Question

1 00000000

1 Byte

```
#include <stdio.h>
int main(){
    char ch;
    for(ch=1;ch;ch++)
        printf("%c\n", ch);
    return 0;
}
```

condition

value of ch

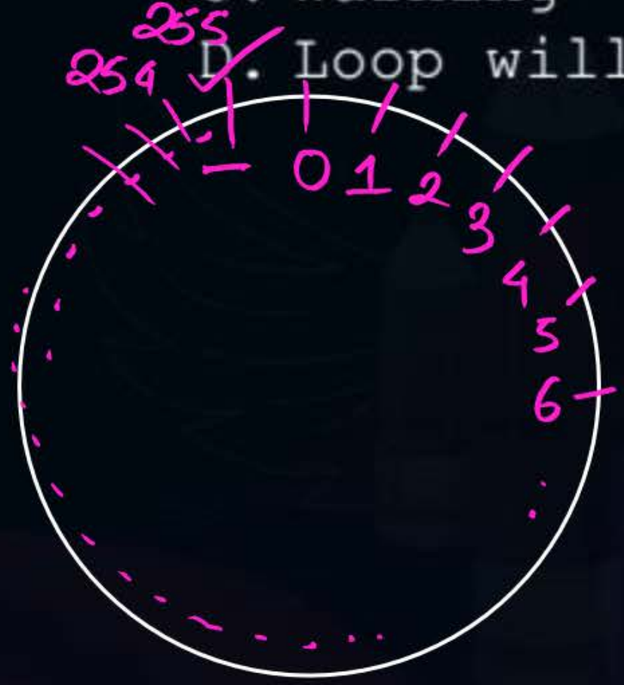
Nonzero/zero

1 Byte

Number of times loop is Executed

A. Compiler error
B. Infinite loop
C. Warning
D. Loop will run 255 time

[D]



1 Byte

unsigned

ranged

Slide



For loop

```
int i;
for( i=10;i>=1 ;i--);
     printf("Name");
```



For loop

```
#include <stdio.h>
int main(){
    int i;
    for(i=1;i<=4;printf("%d\t",i)){
        printf("%d", i);
        i++;
    }
    return 0;
}
```

1<=4 | 2<=4 | 3<=4 | 4<=4 | 5<=4

1 | 2 | 3 | 4 | 5 (false, loop stops)

(i++, 2 | 3 | 4 | 5)

1 2 | 2 3 | 3 4 | 4 5

1 — one time

2 — Every time

3 if condition true

4 if condition is false

Output the program is

A. 12 23 34 45 ✓

B. 11 22 33 44

C. 21 32 43 54

D. error

Slide



For loop

dont apply

UB-LB+1

```
int i;
for( i=1;i<=10 ;i+=2)
    printf("Name");
```

\t = tab character

\n = New Line character

1<=10 - 1 time

3<=10 - 2 times

5<=10 - 3 times

7<=10 - 4 times

9<=10 - 5 times

11<=10 -

Slide



For loop

```
int i;
for( i=1;i<=10 ;i+=2);
     printf("Name");
```

Value of i : 11

No. of times print: 1

for loop will execute



Mathematical Aspect

```
for (i=LB; i<=UB; i=i+K)
    print ("Name");
```

$$\left\lceil \frac{26-15+1}{2} \right\rceil$$

```
for (i=15; i<=26; i=i+2)
```

(6)

15<=26 ✓ 23<=26 ✓
17<=26 ✓ 25<=26 ✓
19<=26 ✓
21<=26 ✓ 27<=26 X

No. of times loop execute

$$\left\lceil \frac{UB-LB+1}{K} \right\rceil$$

Slide



Mathematical Aspect

```
for ( i=2; i<=n; i*=2)
    printf ("Name");
```

```
for ( i=n; i>=2; i/=2)
    printf ("Name");
```

Analysis same

Result

$n >= 2^1$

$n/2^1 >= 2 \Rightarrow n >= 2^2$

$n/2^2 >= 2 \Rightarrow n >= 2^3$

:

$n/2^{k-1} >= 2 \Rightarrow n >= 2^k$

Slide



Mathematical Aspect

```
for ( i=2; i<=n; i*=2)
    printf ("Name");
```

Suppose n=32

No. of times Name printed.

$2<=32$	1
$2^2<=32$	2
$2^3<=32$	3
$2^4<=32$	4
$2^5<=32$ $32<=32$	5
$64<=32$	X

Slide



Mathematical Aspect

```
for ( i=2; i<=n; i*=2)
    printf ("Name");
```

Suppose $n=32$

No. of times Name printed.

$2^k <= n$

$2^2 <= n$

$2^3 <= n$

$\vdots$

$2^k <= n$

$2^k = n$

$\log_2$ both side

$\log_2 2^k = \log_2 n$

$k \log_2 2 = \log_2 n$

$$\boxed{k = \log_2 n}$$

Slide



Mathematical Aspect

```
for (i=2; i<=40; i*=2)
    printf("Name");
```

No. of times

$\lfloor \log_2 n \rfloor$

$2 <= 40$ ✓

$2^2 <= 40$ ✓

$2^3 <= 40$ ✓

$2^4 <= 40$ ✓

$2^5 <= 40$ ✓

$2^6 <= 40$ ✗

Slide



Mathematical Aspect

```
for (i=2; i<=n; i*=2)
    printf("Name")
```

⟶ No. of times printed

$= \lfloor \log_2 n \rfloor$

```
for (i=1; i<=n; i*=2)
    printf("Name")
```

$= \lfloor \log_2 n \rfloor + 1$

Slide



Nested For loop

```
int i,j;
for (i=1; i<=4; i++){
    for (j=1; j<=4; j++)
        printf("%d \t", j);
    printf("\n");
}
```

i=1

→ i=2

→ i=3

→ i=4

→ i=5 - false

j=1	j=2	j=3	j=4
1	2	3	4
1	2	3	4
1	2	3	4
1	2	3	4

Slide



Nested For loop

i = 1 ——— j = 1, j = 2, j = 3, j = 4

i = 2 ——— j = 1, j = 2, j = 3, j = 4

i = 3 ——— j = 1, j = 2, j = 3, j = 4

i = 4 ——— j = 1, j = 2, j = 3, j = 4



Nested For loop

```
int i, j, Sum=0;
for (i=1; i<=10; i++)
    for( j=1; j<=i, j++)
        Sum++;
printf ("%d", sum);
```

What value it will print

Slide



Nested For loop

```
int i, j, Sum=0;
for (i=1; i<=10; i++)
    for( j=1; j<=i, j++)
        Sum++;
printf ("%d", Sum);
```

1+2+3+ · · +10 Sum Incremented

$= \frac{10 \times 11}{2} = 55$

i=1 —— j=1 to 1 — 1
i=2 —— j=1 to 2 — 2
i=3 —— j=1 to 3 — 3
i=4 —— j=1 to 4 — 4
i=5 —— j=1 to 5 — 5
i=6
i=7
i=8
i=9
i=10 —— j=1 to 10 — 10

Slide



GATE 2015

```
#include<stdio.h>
int main(){
    int i, j, k = 0;
    j=2 * 3 / 4 + 2.0 / 5 + 8 / 5;
    k-=--j;
    for (i=0; i<5; i++){
        switch(i+k)
        {
            case 1:
            case 2: printf("\n%d", i+k);
            case 3: printf("\n%d", i+k);
            default: printf("\n%d", i+k);
        }
    }
    return 0;
}
```

GATE 2015

```
#include<stdio.h>
int main()
{
    int i, j, k = 0;
    j=2 * 3 / 4 + 2.0 / 5 + 8 / 5;
    k-=--j;
    for (i=0; i<5; i++)
    {
        switch(i+k)
        {
            case 1:
            case 2: printf("\n%d", i+k);
            case 3: printf("\n%d", i+k);
            default: printf("\n%d", i+k);
        }
    }
    return 0;
}
```

Number of times printf will execute is _____

$j = 2*3/4 + 2.0/5 + 8/5;$

$j = 6/4 + 0.4 + 1$

$j = 1 + 0.4 + 1 = 2.4$

j is integer



$K -= --j \quad : K = K - j$

$K = 0 - 1 = -1$

GATE 2015

```
#include<stdio.h>
int main()
{
    int i, j, k = 0;
    j=2 * 3 / 4 + 2.0 / 5 + 8 / 5;
    k-=--j;
    for (i=0; i<5; i++)
    {
        switch(i+k)
        {
            case 1:
            case 2: printf("\n%d", i+k);
            case 3: printf("\n%d", i+k);
            default: printf("\n%d", i+k);
        }
    }
    return 0;
}
```

Number of times printf will execute is _____

k = -1

i	Switch(i+k)	No. of printf
i = 0	-1	→ 1
i = 1	0	→ 1
i = 2	1	→ 3
i = 3	2	→ 3
i = 4	3	→ 2
		10



For loop

$\left\lceil \frac{100-1+1}{3} \right\rceil = \lceil 33.33 \rceil = 34$

The number of times loop executed is

```
#include<stdio.h>
int main(){
      int i,count=0;
      for (i=1; i<=100; ){
            printf("\nI am a good Student");
            i = i+3;
      }
      return 0 ;
}
```

$\left\lceil \frac{100}{3} \right\rceil = \lceil 33.33 \rceil = 34$ times

$i<=10$ — $1<=10$, $4<=10$, $7<=10$, $10<=10$ } 4 times

$\left\lceil \frac{10}{3} \right\rceil = \lceil 3.33 \rceil = 4$

$i<=11$ — 4 times — $1<=10$, $4<=10$, $7<=10$, $10<=10$

$i<=12$ — 4 times

$i<=13$ — 5 times — $13<=10$ X

$\left\lceil \frac{13}{4} \right\rceil$: 5 times $\left\lceil \frac{11}{3} \right\rceil = \lceil 3.6 \rceil = 4$

(A) 33
(B) 34 ✓
(C) 100
(D) 10

Slide

n^{th} term - $a+(n-1)d$

$a = 1$

$n = ?$

$d = 3$

$100 = 1+(n-1)3$

$(n-1)3 = 99$

$n-1 = 33$

$n = 34$



For loop

What is the output of the program

```
#include<stdio.h>
int main(){
    int i,sum=0;
     for (i=1; i<=100;i++){
          sum= sum+i;
     }
     printf("%d",sum);
     return 0 ;
}
```

(A) 100
(B) 5050
(C) 10100
(D) 10000

Slide

i=1
i=2
i=3
i=4
i=5

:

i=100

Sum: Sum+i	Sum
	0+1
	0+1+2
	0+1+2+3
	0+1+2+3+4
	0+1+2+3+4+5

1+2+3+ - - - +10

Sum of n Natural No = $\frac{n(n+1)}{2}$

$50\,\frac{100\times101}{2} = 5050$



2 mins Summary

Topic for loop

Topic Nested for loop

Topic Mathematical aspect

Topic

Topic

Slide

THANK - YOU